AI OPEN
DATA CENTER:
Power
Distribution



Using the Full Low Voltage Direct Current Range up to 1500V for Data Centers

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Low Voltage Direct Current Range up to 1500V

Low Voltage Range

- for AC up to 1000V (rms)
- for DC up to 1500V

International Standard Voltages 350V, 700V and 1400V

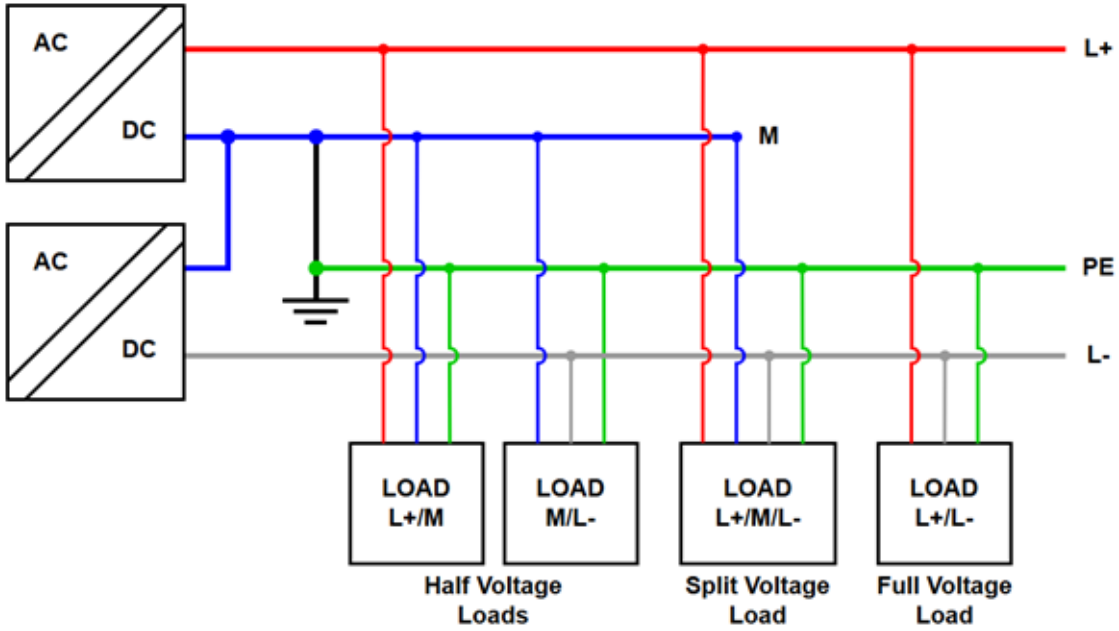
Equivalent in data center context 400V, 800V and 1500V

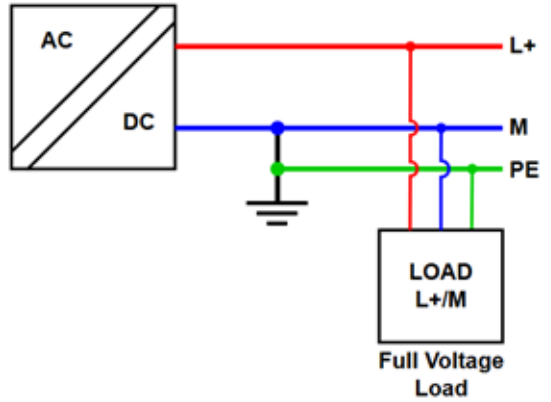
Typically we use:

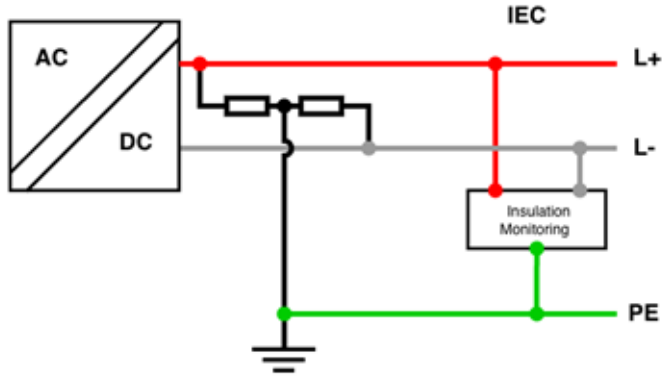
- Lower voltages for low power
- Higher voltages for higher power

Source: OCP White Paper: Data Center Facility - Low Voltage Direct Current Power Distribution V1.0

Polarities & System Earthing







Combinations of Polarities, System Earthing and Voltages:

- What is the impact of system choices on components?
- What kind of systems can a component be made compatible with?

Impact of Voltage and System Choices on Material, Losses and Semiconductor Breakers

Nominal Voltage (equivalent) (V)	**350/700** (400/800)		**700** (800)				**700/1400** (750/1500)		**1400** (1500)			
Polarities	**Bipolar**				**Unipolar**		**Bipolar**				**Unipolar**	
Earthing	**TN-S**	**IT**	**TN-S**	**IT**	**TN-S**	**IT**	**TN-S**	**IT**	**TN-S**	**IT**	**TN-S**	**IT**

Mt. Diablo 0.7.0 Spec March 2026

NVIDIA Architecture white paper 2025

Google Presentation OCP Summit 2025

Impact of Voltage and System Choices on Material, Losses and Semiconductor Breakers

Nominal Voltage (equivalent) (V)	**350/700** (400/800)		**700** (800)				**700/1400** (750/1500)		**1400** (1500)			
Polarities	**Bipolar**				**Unipolar**		**Bipolar**				**Unipolar**	
Earthing	**TN-S**	**IT**	**TN-S**	**IT**	**TN-S**	**IT**	**TN-S**	**IT**	**TN-S**	**IT**	**TN-S**	**IT**
Active Wires	2-3	2-3	2	2	2	2	2-3	2-3	2	2	2	2
Relative Active Wire Material	2-3	2-3	2	2	2	2	1-1.5	1-1.5	1	1	1	1
Relative Cable Losses for same balanced Power	1	1	1	1	1	1	0.5	0.5	0.5	0.5	0.5	0.5
Relative Current for same Balanced Power	1	1	1	1	1	1	0.5	0.5	0.5	0.5	0.5	0.5
Min. # of Overcurrent Protected Poles	2	3[h]	2	2[h]	1	2[h]	2	3[h]	2	2[h]	1	2[h]
Relative Voltage Semiconductor Circuit Breaker[i]	0.5	1	1	1	1	1	1	2	2	2	2	2
Semiconductor Circuit Breaker FoM[j]	1	3	2	2	1	2	1	3	2	2	1	2

Lower numbers are better

Impact of Voltage and System Choices on Density

Nominal Voltage (equivalent) (V)	**350/700** (400/800)		**700** (800)				**700/1400** (750/1500)		**1400** (1500)			
Polarities	**Bipolar**				**Unipolar**		**Bipolar**				**Unipolar**	
Earthing	**TN-S**	**IT**	**TN-S**	**IT**	**TN-S**	**IT**	**TN-S**	**IT**	**TN-S**	**IT**	**TN-S**	**IT**
Design steady state nominal line-to-earth voltage (V)	350	700	350	700	700	700	700	1400	700	1400	1400	1400
Impulse Withstand Voltage, OVC I (kV)[a]	1.5	2.5	1.5	2.5	2.5	2.5	2.5	4	2.5	4	4	4
Impulse Withstand Voltage, OVC II (kV)[a]	2.5	4	2.5	4	4	4	4	6	4	6	6	6
Impulse Withstand Voltage, OVC III (kV)[a]	4	6	4	6	6	6	6	8	6	8	8	8
Impulse Withstand Voltage, OVC IV (kV)[a]	6	8	6	8	8	8	8	12	8	12	12	12
Clearance to PE, OVC II (mm)[b]	1.5	3	1.5	3	3	3	3	5.5	3	5.5	5.5	5.5
Clearance to PE, OVC III (mm)[b]	3	5.5	3	5.5	5.5	5.5	5.5	8	5.5	8	8	8
Clearance to PE, OVC IV (mm)[b]	5.5	8	5.5	8	8	8	8	14	8	14	14	14
Creepage to PE, PD2, for Mat Grp 1 (mm)[c]	2.5	5	2.5	5	5	5	5	8	5	8	8	8
Creepage to PE, PD2, Mat Grp 3 (mm)[c]	5	10	5	10	10	10	10	16	10	16	16	16
Creepage to PE, PD3, Mat Grp 3 (mm)[c]	8	16	8	16	16	16	16	25	16	25	25	25

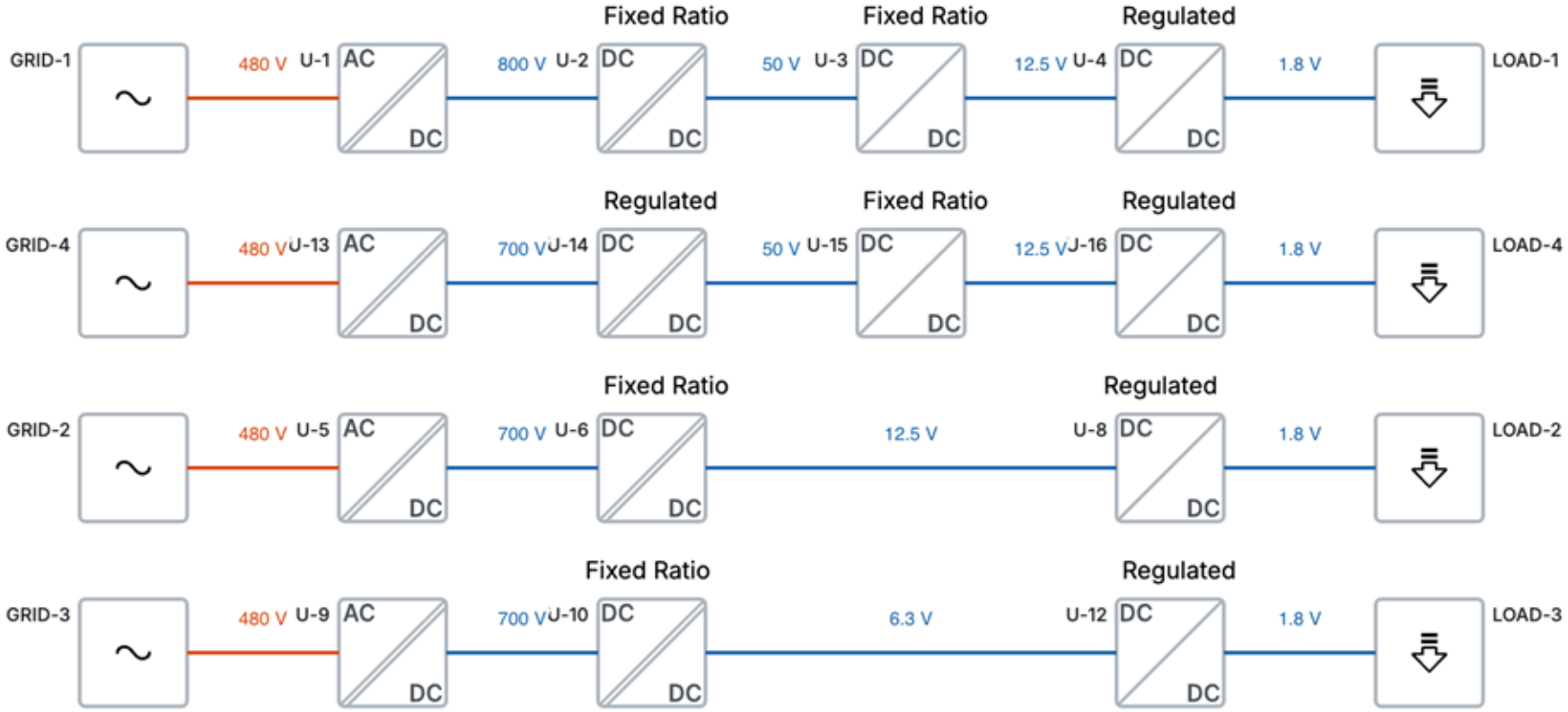
Lower numbers are better

Line-to-Earth Voltage and Compatibility

Nominal Voltage to Earth	350/700 (400/800)		700 (800)				700/1400 (750/1500)		1400 (1500)			
	Bipolar				Unipolar		Bipolar				Unipolar	
	TN-S	IT	TN-S	IT	TN-S	IT	TN-S	IT	TN-S	IT	TN-S	IT
+1400												
+700												
+350												
-350												
-700												
-1400												
Compatibility:												
350V	x	x										
700V	**x**	x	**x**	**x**	**x**	**x**	**x**	x				
1400V							x	x	x	x	x	x

Prepare 800V Equipment for 700/1400V

- Allow operation at 700V (lower lower limit)
- Ensure setting for sources to limit to 750V



Designing for 700V and 800V

In rack voltage ratio examples

700V

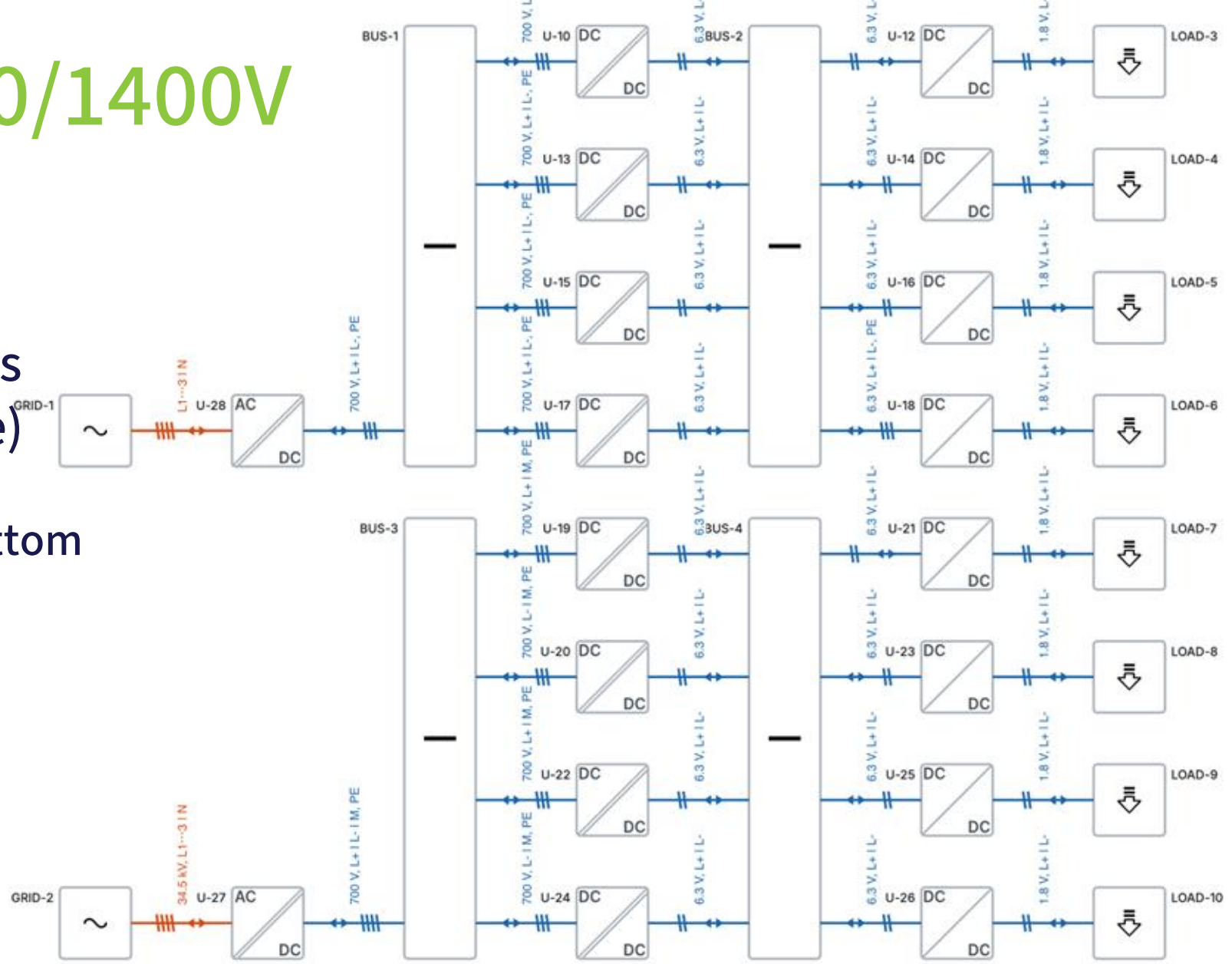
	120.00%	**840**	60.00	7.50
	114.29%	800	57.14	7.14
	108.57%	760	54.29	6.79
Nominal	**100.00%**	**700**	**50.00**	**6.25**
	91.43%	640	45.71	5.71
	90.00%	630	45.00	5.63
	88.57%	620	44.29	5.54
Ratio		1	14	112

800V

	105.00%	**840**	52.50	6.56
	103.75%	830	51.88	6.48
	102.50%	820	51.25	6.41
Nominal	**100.00%**	**800**	**50.00**	**6.25**
	95.00%	760	47.50	5.94
	87.50%	700	43.75	5.47
	80.00%	640	40.00	5.00
Ratio		1	16	128

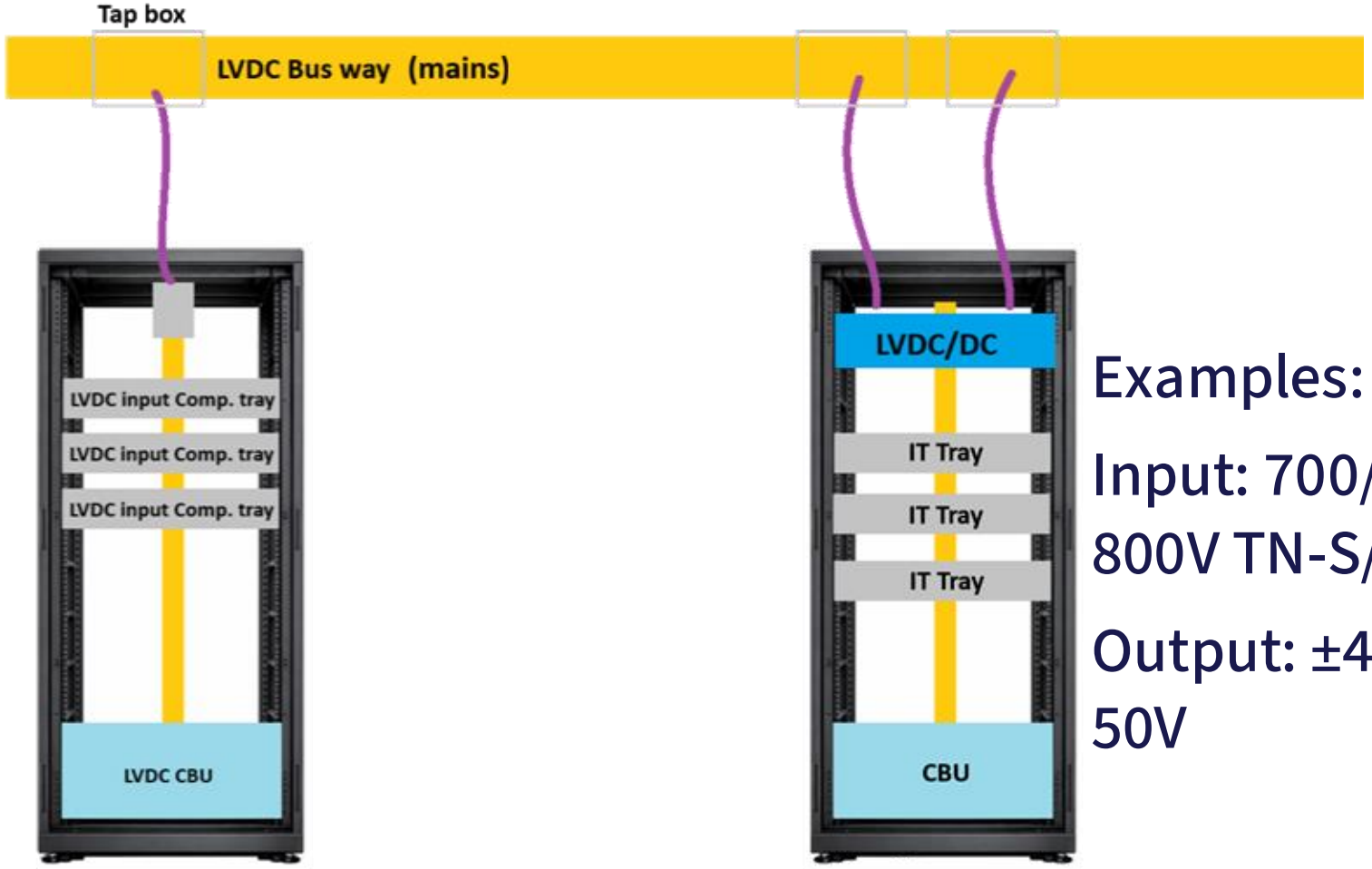
Connecting to 700/1400V

1. Alternating 700V Racks
2. Alternating 700V Servers
3. Alternating DC/DC supplies inside server (right picture)
   a. 700V input on top
   b. 700/1400V input on the bottom



DC/DC Conversion Shelf

Examples:
Input:
700/1400V TN-S or
800V TN-S/IT



Examples:

Input: 700/1400V TN-S or 800V TN-S/IT

Output: ±400V/800V or 50V

LVDC-Native IT tray

Universal conversion shelf

Call to Action

1. 800V devices designed for IT grounding or unipolar TN-S grounded systems automatically meet the creepage and clearance requirements for 700/1400V TN-S.
2. Design 800V devices such that they can operate under 750V as well for use in 700/1400V TN-S systems (including option to limit source max voltage to 750V)
3. Wider operation voltage range beneficial anyway in larger DC grids and fault ride through

Get involved in the project:
https://www.opencompute.org/projects/power-distribution

OCP White Paper: Data Center Facility - Low Voltage Direct Current Power Distribution V1.1.0
https://www.opencompute.org/documents/dcf-power-distribution-lvdc-white-paper-version-1-1-0-pdf

Open Discussion



Footnotes

a) Impulse Withstand Voltage(Uw, Uimp): IEC 60364-4-44, currently under revision and a new table with DC values is proposed. The DC voltages are proposed as 1.5 times the AC rms voltage.

b) Clearance: IEC 60664-1, Table F.2, Case A inhomogeneous field. Values are for Basic Insulation (BI). OVC IV included for SST applications. Tables that define clearance distances in IEC 62368-1 also make reference to IEC 60664-1. Other relevant product standards (incl. NEC and UL) and their relation to the values in the table is shown below in section 6.3.1.1.

c) Creepage: IEC 60664-1, Table F.5, values for pollution degree 2 and 3 are shown (Note: use of ribs can reduce PD3 requirement), Material Group (Mat Grp) 1 for PCB design, Material Group 3 for Assembly. Values based on Vrms at 500V, 1000V, 1600V for 350V, 700V and 1400V, respectively. Values are for Basic Insulation (BI). Tables that define creepage distances in IEC 62368-1 also make reference to IEC 60664-1. Other relevant product standards (incl. NEC and UL) and their relation to the values in the table is shown below in section 6.3.1.1.

d) Recommendations for insulation coordination based on IEC 62477-1,-2, for HF TX between primary and secondary. (RI) Reinforced Insulation, (BI) Basic Insulation. In general, if it is accessible to humans it must be RI, if a human cannot access - it could be BI and if it can be tripped fast (RCD) - it could be BI. For a TN-S DC grid and there is SSCB, you could define this as BI with or without human accessibility. Topic is still under discussion and SST application based standards are still under development for this topic. See section 6.3.1.2 for more information.

e) Requires 1x the rated voltage and corresponding electrical spacing of the device from earth. See section below section 6.3.2 Combination Table Analysis: Device Compatibility for detailed description and legend, see Table 7.2for breakdown of x and (x).

f) Requires 2x the rated voltage and corresponding electrical spacing of the device from earth. See Table 11 for breakdown of x and (x).

g) With (partially) distributed M. For some characteristics this results in a range from having M distributed everywhere, to only having it in some circuits, and to no distribution of M but just usage in protection cabinets.

h) NEC 240.15(A) requires all ungrounded poles (regardless of TN-S or IT system type) to be protected, this is similar for IT requirements in references such as the IEC 60364-4-43 and NEN 1010. It is considered that IT earthed systems are designed for continued operation (for a limited time) after the first ground fault.

i) Semiconductor voltage example reference: 0.5=600V, 1=1200V, 2=2400V). Semiconductor selection (in a protection device) may vary by design. See section below titled section 6.3.4 Combination Table Analysis: Impact on Protection Devices for more information.

j) This Figure of Merit (FoM) quantifies the total semiconductor "effort" or cost required to protect a system. It is calculated by the multiplication of relative Semiconductor Circuit Breaker Voltage, Relative Current for same Power and Number of Protected Poles. For the current, the balanced case is considered in Bipolar systems with M. It is assumed that unbalance typically only occurs under lower load conditions.